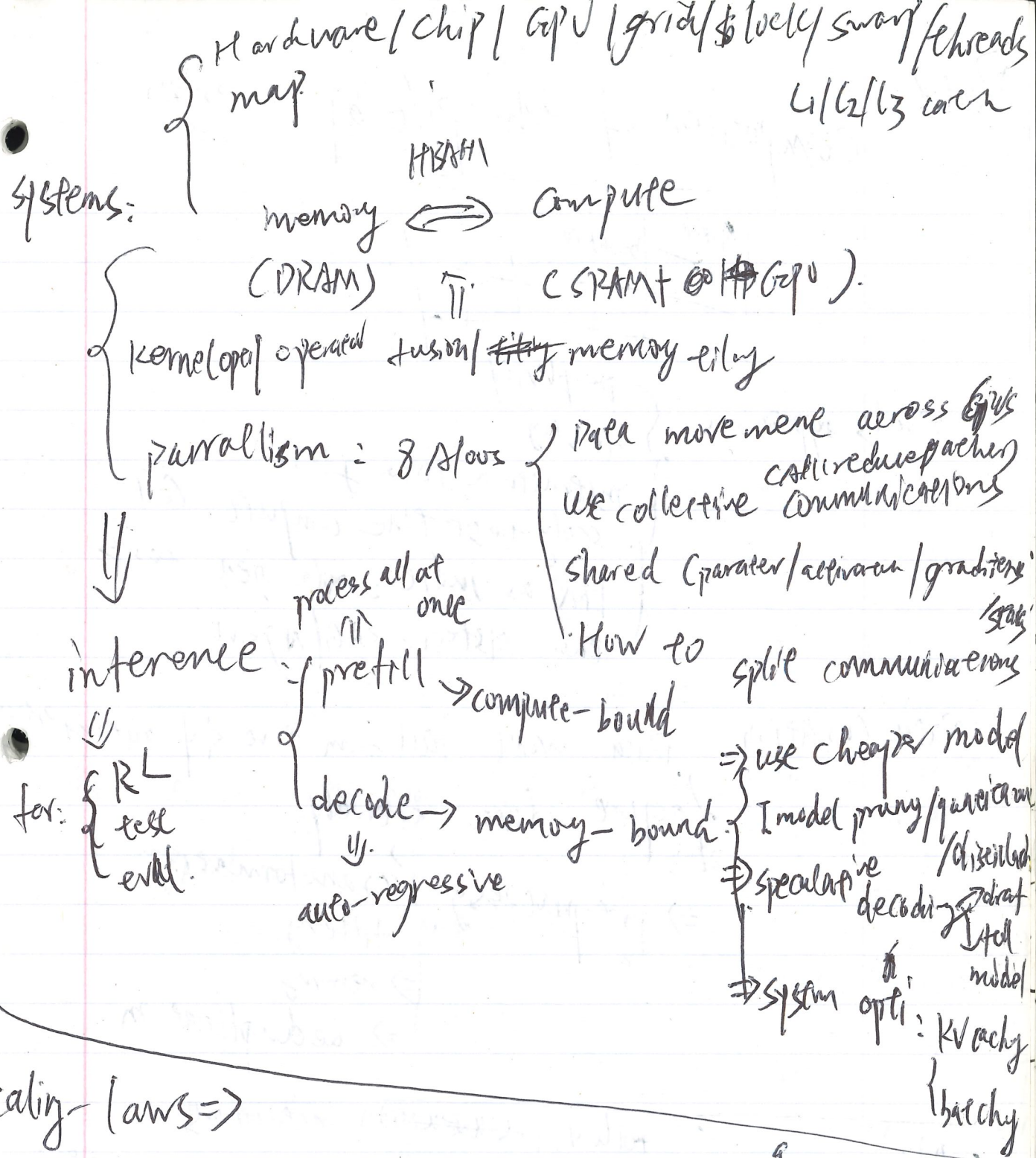


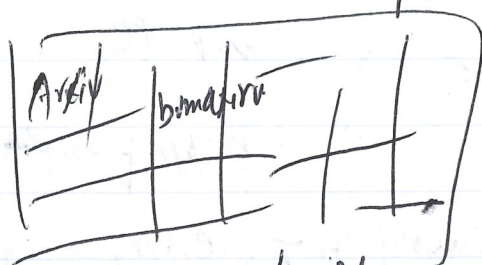


Scaling laws $\Rightarrow$

Model size $\times 20 =$ Num of tokens for training: (1.4B models requires 28B training tokens)

Assignment: implement baby version

Data: composition of the pile by category



evals approaches: { perplexity
testng.
interaction follow.
scalp test-time compute: CoT
LM-as-judge: eval gen ~~tasks~~
Full-system: RAG/Agent

Data Curation: Data won't fall from the sky automatically

acquire data actively

=> pre-processing { transformation
filtering
removing
deduplication

Alignment: { follow ~~instruction~~ instructions
Tune style (format, length, tone, etc)
incorporate safety (refusal to answer questions)

step 1: SFT
step 2: HRLF

=> Verifiers { format = (code/math)
learned verifiers => LM-as-judge